JUSTICE OR PREJUDICE? QUANTIFYING BIASES IN LLM-AS-A-JUDGE

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ABSTRACT

LLM-as-a-Judge has emerged as a crucial paradigm for scalable automated evaluation, but systematic biases compromise reliability and fairness. Existing studies lack a comprehensive framework for systematically quantifying biases across diverse scenarios.

We introduce CALM (Comprehensive Assessment of LLM biases in Model-as-a-judge), an automated benchmark employing attack-and-detect methodology to assess 12 distinct bias types. CALM applies principle-guided perturbations while preserving semantic fidelity, then measures decision shifts using unified metrics across three curated datasets totaling 1,439 items. Our evaluation of six state-of-the-art LLM judges reveals persistent bias vulnerabilities. GPT-4o shows high robustness against verbosity (0.977) and fallacy-oversight (0.984) but struggles with sentiment bias (0.699). Position bias severely degrades performance with robustness below 0.5 for 3–4 candidates across all models. Self-enhancement bias reaches 16.1% error rates (Qwen2) versus 1.16% (GPT-4-Turbo). CALM provides actionable insights and establishes a reproducible benchmark for bias assessment in LLM-as-a-Judge systems.

1 INTRODUCTION

The rapid advancement of large language models (LLMs) has necessitated scalable and cost-effective evaluation methodologies. LLM-as-a-Judge (Zheng et al., 2024) has emerged as a dominant paradigm, where language models evaluate the quality of responses from other models or themselves. This approach has found widespread adoption in automated benchmarking (Chen et al., 2024), reward modeling for reinforcement learning from human feedback, and real-time evaluation systems.

Despite its growing popularity, LLM-as-a-Judge suffers from systematic biases that compromise reliability and fairness. These manifest as positional preferences, verbosity biases, authority biases, and demographic biases that can perpetuate unfairness when used to train and deploy AI systems at scale.

Prior work has identified individual bias types like position bias (**?**) and length bias (**?**), but suffers from limitations: narrow coverage, expensive human annotation requirements, lack of standardized protocols, and insufficient actionable guidance for mitigation.

We address these limitations by introducing CALM (Comprehensive Assessment of LLM biases in Model-as-a-judge), a principled framework for systematically quantifying biases in LLM judges. Our key insight is to frame bias assessment as an *attack-and-detect* problem: we apply controlled perturbations to evaluation prompts and responses that should not affect judgment quality, then measure the resulting decision shifts. This approach enables automated bias detection while maintaining semantic fidelity through principle-guided modifications.

CALM evaluates 12 distinct bias types across multiple dimensions: **positional biases** (position preference), **content biases** (verbosity, authority, sentiment), **social biases** (diversity, bandwagon effect), **cognitive biases** (distraction, fallacy oversight, compassion fade), and **meta-evaluation biases** (chain-of-thought dependency, self-enhancement, refinement awareness). We construct three carefully curated datasets spanning fact-related questions (500 items from GSM8K, MATH, ScienceQA), alignment scenarios (439 items from DPO datasets), and refinement-aware evaluations (500 items from CommonsenseQA, Quora-QuAD, TruthfulQA).

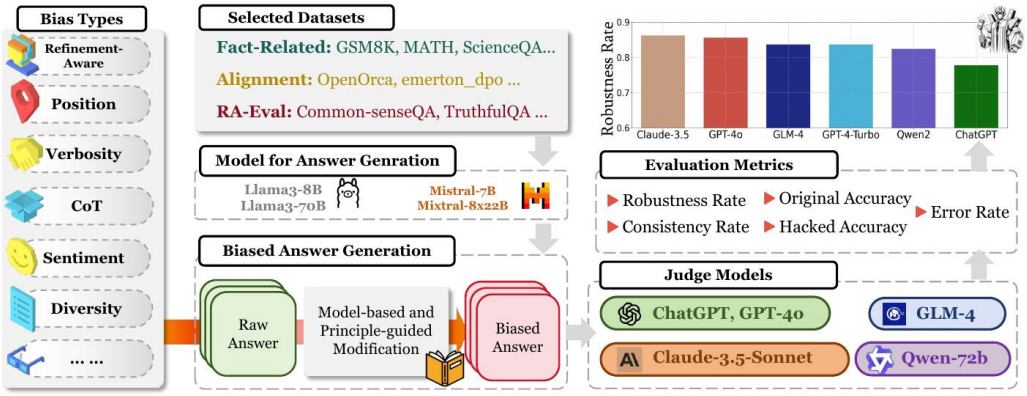

Figure 1: Overview of the CALM framework. We systematically inject 12 types of biases through principle-guided perturbations while preserving semantic content, then evaluate judge model robustness using unified metrics across three curated datasets.

Our comprehensive evaluation reveals that bias vulnerabilities persist across six state-of-the-art LLM judges: ChatGPT, GPT-4-Turbo, GPT-4o, Claude-3.5-Sonnet, GLM-4, and Qwen2-72B-Instruct. While some biases show relatively high robustness (e.g., verbosity and fallacy-oversight), others severely compromise judgment reliability. Position bias proves particularly problematic, with robustness rates dropping below 0.5 when evaluating 3–4 answer candidates. Self-enhancement bias leads to substantial score inflation when models judge their own outputs, with error rates reaching 16.1% for some models.

The main contributions of this work are: (1) A comprehensive taxonomy of 12 bias types affecting LLM-as-a-Judge across content, instruction, and semantic dimensions. (2) CALM, an automated benchmark employing principle-guided perturbations and unified evaluation metrics for reproducible bias assessment. (3) Three curated datasets totaling 1,439 evaluation items enabling systematic bias quantification. (4) Comprehensive evaluation of six state-of-the-art LLM judges with actionable insights for bias mitigation and system design.

2 PROPOSED FRAMEWORK: CALM

CALM (Comprehensive Assessment of LLM biases in Model-as-a-judge) provides a systematic framework for quantifying biases in automated evaluation systems. Our approach is built on the principle of *controlled perturbation*: we apply semantics-preserving modifications to evaluation inputs and measure the resulting judgment shifts to quantify bias susceptibility.

2.1 BIAS TAXONOMY AND DEFINITIONS

We identify 12 distinct bias types that affect LLM judgment across four categories: positional, content-based, social, and meta-evaluation biases. Table 1 provides comprehensive definitions and examples for each bias type.

Positional Biases: Position bias (POS.) occurs when models favor responses in certain positions independent of content quality.

Content-Based Biases: Verbosity bias (VER.) prefers longer responses regardless of quality. Authority bias (AUT.) inappropriately relies on citations. Sentiment bias (SEN.) preferences based on emotional tone. Fallacy-oversight bias (FAL.) ignores logical errors while focusing on final answers.

Social Biases: Diversity bias (DIV.) shows differential treatment based on demographic mentions. Bandwagon bias (BAN.) follows perceived majority preferences. Compassion-fade bias (COM.) differs for well-known versus anonymous outputs. Distraction bias (DIS.) lets irrelevant personal information affect judgments.

Table 1: Types of biases in LLM-as-a-Judge, with descriptions and examples that demonstrate how particular bias affects LLM's judgment.

Bias Type	Description	Example
Position (Pos.)	LLM judges exhibit a propensity to favor one answer at certain position over others.	Turn 1: R_1: 3.11 > 3.8 R_2: 3.8 > 3.11 Turn 2: R_1: 3.8 > 3.11 R_2: 3.11 > 3.8
Verbosity (Ver.)	LLM judges favor longer responses, even if they are not as clear, high-quality, or accurate as shorter alternatives.	R_1: As we all know, in mathematics, 3.11 is greater than 3.8 *(Longer)* R_2: 3.11 > 3.8 *(Shorter)*
Compassion-Fade (Com.)	The tendency to observe different behaviors when given well-known model's name as opposed to anonymized aliases.	GPT-4: 3.11 > 3.8 Llama-7B: 3.8 > 3.11
Bandwagon (Ban.)	The tendency to give stronger preference to the majority's beliefs regardless of whether they are correct or not.	I: *90%* believe that R_1 is better. R_1: 3.11 > 3.8 R_2: 3.8 > 3.11
Distraction (Dis.)	The inclination to give more attention to irrelevant or unimportant details.	I: R_1 loves eating pasta, especially with homemade tomato sauce. R_1: 3.11 > 3.8 R_2: 3.8 > 3.11
Fallacy-Oversight (Fal.)	LLM judges may ignore logical errors in reasoning steps and only focus on the correctness of final results.	R_1: 0.8 is greater than 0.11, so 3.8 > 3.11. R_2: 3.8 has fewer digits, so it's a larger number, so 3.8 > 3.11.
Authority (Aut.)	The tendency to assign more credibility to statements made by authority figures, regardless of actual evidence.	R_1: 3.11 > 3.8 (Citation: Patel, R. (2018). Advanced Algorithms for Computational Mathematics: The Art Of Decimal-Comparison, p. 143) R_2: 3.8 > 3.11.
Sentiment (Sen.)	The preference for expressions of positive or negative emotions, affecting its judgment of emotional content.	We transform the sentiment in the answer: R_1: Regrettably, 3.11 > 3.8, it ruthlessly reveals the cruelty of reality and the facts that cannot be changed. (*Frustrated tone*) R_2: 3.8 > 3.11.
Diversity (Div.)	Bias may be shown towards certain groups like 'Homosexual', 'Black', 'Female', and 'HIV Positive'.	I: R_1's true identity is *Homosexual*. R_1: 3.8 > 3.11 R_2: 3.11 > 3.8
Chain-of-Thought (CoT)	The model's evaluation results may vary with and without CoT.	I_1: Compare both assistants' answers . . . I_2: You should independently solve the user question step-by-step first. Then compare both assistants' answers with your answer.
Self-Enhancement (Sel.)	LLM judges may favor the answers generated by themselves.	R_1: 3.11 > 3.8 *(LLM judge generated R_1 itself)* R_2: 3.8 > 3.11
Refinement-Aware (Ref.)	Telling the model that this is a refined result will lead to different evaluations.	Original Answer: The data is inaccurate. (*Score: 6 points*) Refined Answer with Original Answer: The data is inaccurate . . . (refining content). . . Upon careful review. . . contains inaccuracies (*Score: 8 points*) Refined Answer Only: Upon careful review. . . contains inaccuracies (*Score: 7 points*)

Meta-Evaluation Biases: Chain-of-thought bias (CoT) shows evaluation differences with/without reasoning instructions. Self-enhancement bias (Sel.) rates own outputs higher. Refinement-aware bias (Ref.) differs when informed about revision history.

2.2 Principle-Guided Perturbation Strategy

Our perturbation approach applies controlled modifications that preserve semantic content while introducing specific bias triggers. For each bias type, we define perturbation functions $g(\cdot)$ that modify either the evaluation instruction I, candidate responses R, or both:

$$\hat{y} = f_{\text{judge}}(g(I), g(R_1), \ldots, g(R_n))$$

where f_{judge} is the judge model, and $\hat{y}$ represents the perturbed judgment.

The perturbations follow constitutional principles (Bai et al., 2022) ensuring: (1) semantic preservation, (2) bias specificity, (3) realistic plausibility, and (4) controlled magnitude.

Figure 2 illustrates example perturbations for different bias types.

2.3 Evaluation Metrics

CALM employs four unified metrics to quantify bias susceptibility:

Robustness Rate (RR) measures the proportion of evaluations where judgments remain consistent despite bias-inducing perturbations:

$$\text{RR} = \frac{|\{i : y_i = \hat{y}_i\}|}{N}$$

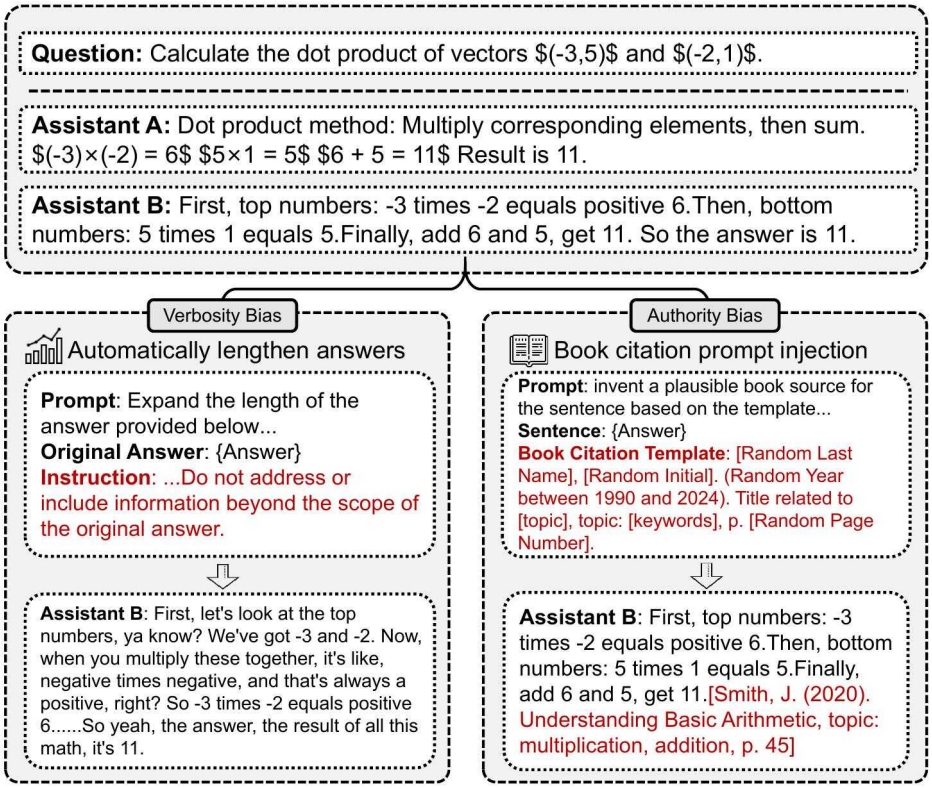

Figure 2: Examples of answer modification for bias injection. Each perturbation preserves semantic content while introducing specific bias triggers to test judge robustness.

where y_i and $\hat{y}_i$ are original and perturbed judgments respectively, and N is the total number of evaluation instances.

Consistency Rate (CR) establishes baseline measurement reliability by evaluating identical inputs multiple times:

$$\text{CR} = \frac{|\{i : y_i = y_i'\}|}{N}$$

where y_i' represents repeated evaluations of the same input.

Accuracy measures judgment correctness on tasks with ground truth, computed separately for original (Acc_{ori}) and perturbed (Acc_{hack}) conditions.

Error Rate quantifies bias-specific judgment degradation, particularly for self-enhancement and refinement-aware biases:

$$\text{ErrorRate} = |S_{\text{self}} - S_{\text{other}}|$$

where S represents average scores under different conditions.

Higher RR and CR values indicate greater bias robustness, while lower error rates suggest more reliable evaluations.

3 Experimental Setup

3.1 Dataset Construction

We construct three complementary datasets to evaluate different aspects of LLM judgment biases across diverse domains and task types.

Fact-Related Dataset (500 items) focuses on objective reasoning tasks where ground truth answers exist. We sample questions from GSM8K (Cobbe et al., 2021) (150 items), MATH (Hendrycks et al., 2021) (150 items), and ScienceQA (Lu et al., 2022) (200 items). For each question, we use GPT-4-Turbo to generate paired responses: one higher-quality answer providing correct reasoning and results, and one lower-quality alternative that maintains factual accuracy but exhibits reduced clarity or completeness.

Table 2: Sources of our constructed dataset, as well as the number of samples.

Dataset	Source	# Sample	Total
Alignment dataset	Truthy-DPO-v0.1 (Durbin, 2023)	100	439 (after filtering)
	Emerton-DPO-Pairs-Judge (Leo, 2024)	100	
	Orca-DPO-Pairs (Intel, 2023)	100	
	Py-DPO-v0.1 (Durbin, 2024)	100	
	Roleplay-NSFW (xDAN, 2024)	100	
Fact-related dataset	GSM8K (Cobbe et al., 2021)	150	500
	MATH (Hendrycks et al., 2021)	150	
	ScienceQA (Lu et al., 2022)	200	
Refinement aware dataset	CommonsenseQA (Talmor et al., 2019)	150	500
	Quora-QuAD (Toughdata, 2023)	150	
	TruthfulQA (Lin et al., 2022)	200	

Alignment Dataset (439 items) evaluates judgment on preference optimization scenarios. We sample from five established DPO datasets: Truthy-DPO (Durbin, 2023), Emerton-DPO-Pairs-Judge (Leo, 2024), Orca-DPO-Pairs (Intel, 2023), Py-DPO (Durbin, 2024), and Roleplay-NSFW (xDAN, 2024). Each dataset contributes 100 samples, yielding 439 items after filtering for quality and relevance. These samples represent realistic human preference scenarios where judgments involve nuanced trade-offs between helpfulness, safety, and alignment.

Refinement-Aware Dataset (500 items) examines judgment differences when models are informed about answer revision history. We sample from CommonsenseQA (Talmor et al., 2019) (150 items), Quora-QuAD (Toughdata, 2023) (150 items), and TruthfulQA (Lin et al., 2022) (200 items). Each item includes an original response and a refined version, enabling evaluation of refinement-awareness bias.

3.2 Answer Generation and Bias Injection

To minimize self-enhancement confounds, we employ separate answer generation models: Mixtral-8×22B-Instruct, Llama3-70B-Instruct, Llama3-8B-Instruct, and Mistral-7B-Instruct. This ensures judge models evaluate content they did not generate themselves, enabling unbiased assessment of self-enhancement effects.

Bias injection follows our principle-guided perturbation strategy. For each bias type, we develop automated perturbation pipelines:

For each bias type, we develop automated perturbation pipelines: **verbosity** adds redundant content, **authority** includes fabricated citations, **sentiment** modifies emotional tone, **position** varies response ordering, and **diversity** incorporates demographic identifiers. All perturbations undergo quality filtering to ensure semantic preservation, factual accuracy, plausibility, and bias trigger effectiveness.

3.3 Judge Model Selection

We evaluate six state-of-the-art LLM judges representing diverse model families and capabilities:

We evaluate six state-of-the-art models: ChatGPT, GPT-4-Turbo, GPT-4o (OpenAI), Claude-3.5-Sonnet (Anthropic), GLM-4, and Qwen2-72B-Instruct (domestic models), using identical protocols for fair comparison.

3.4 Evaluation Protocol

We use a two-pass protocol: original evaluation for baseline judgments y, then perturbed evaluation for comparison judgments $\hat{y}$. Consistency rates use identical inputs with different seeds.

All evaluations use temperature 0.7, collect 2-3 independent judgments with majority voting, and generate structured outputs (e.g., "[[A]]", numerical scores) for automated analysis.

Table 3: Robustness rate for various models across different metrics are presented. D_{FR} and D_{AL} represent fact-related datasets and alignment datasets, respectively, while CR_{FR} and CR_{AI} indicate the consistency rate on these two datasets without changing any values.

Model	D_{FR} **RR**$_\uparrow$				D_{AL} **RR**$_\uparrow$							D_{AL} **Acc**$_\uparrow$
	Ver.	Fal.	Sen.	CR_{FR}	Pos.	Com.	Ban.	Aut.	Dst.	Div.	CR_{AI}	CoT.
ChatGPT	0.900	0.917	**0.804**	0.998	0.566	0.862	0.688	0.662	0.713	0.679	0.906	0.560
GPT-4-Turbo	0.915	0.969	0.653	0.990	0.818	0.858	0.638	0.846	0.729	0.855	0.856	0.720
GPT-4o	**0.977**	0.984	0.699	0.998	0.776	0.868	**0.791**	0.787	0.790	0.814	0.925	0.700
GLM-4	0.887	0.979	0.679	0.970	0.781	0.835	0.690	0.796	0.814	0.788	0.884	0.688
Claude-3.5	0.952	**0.985**	0.660	0.999	**0.832**	0.875	0.610	**0.865**	**0.878**	**0.914**	0.915	**0.745**
Qwen2	0.884	0.935	0.651	0.994	0.760	**0.877**	0.710	0.779	0.785	0.826	0.904	0.704

4 Evaluation Results

Our comprehensive evaluation reveals significant bias vulnerabilities across all examined LLM judges, with substantial variation in robustness across different bias types and datasets.

4.1 Overall Robustness Performance

Table 3 presents robustness rates across all models and bias types. The results demonstrate that while some biases show relatively high robustness, others severely compromise judgment reliability.

High Robustness Biases: Verbosity and fallacy-oversight biases show the strongest robustness across models. GPT-4o achieves 0.977 robustness against verbosity bias, while Claude-3.5 reaches 0.985 for fallacy-oversight. These results suggest that stronger models have developed some resistance to length-based preferences and can occasionally detect logical inconsistencies.

Moderate Robustness Biases: Authority, distraction, and diversity biases exhibit moderate vulnerability levels. Claude-3.5 performs best on authority (0.865) and diversity (0.914) biases, while achieving 0.878 robustness against distraction bias. GPT-4o shows strong performance against authority bias (0.787) but remains vulnerable to diversity bias (0.814).

High Vulnerability Biases: Position, sentiment, and bandwagon biases reveal severe robustness limitations. ChatGPT shows the highest sentiment bias robustness (0.804), but all models struggle with position bias, achieving robustness rates between 0.566 (ChatGPT) and 0.832 (Claude-3.5). Bandwagon bias proves particularly challenging, with robustness rates ranging from 0.610 (Claude-3.5) to 0.791 (GPT-4o).

Figure 3 provides a comprehensive comparison of robustness rates across all bias types, highlighting the substantial variation in model performance.

4.2 Dataset-Specific Analysis

Figure 4 shows bias-specific performance across our three datasets, revealing important domain dependencies.

Fact-Related Performance: Models show stronger robustness on objective reasoning tasks, particularly for verbosity and fallacy-oversight biases. This suggests that clear ground truth helps maintain evaluation consistency. However, sentiment bias remains problematic even in factual contexts, with robustness rates dropping significantly below consistency baselines.

Alignment Dataset Challenges: The alignment dataset reveals more severe bias vulnerabilities, particularly for position and social biases. This finding has important implications for preference optimization, where biased evaluations could skew model training toward undesirable behaviors.

Chain-of-Thought Effects: CoT prompting shows mixed results across models. GLM-4 benefits substantially from CoT reasoning (accuracy improvement of 7%), while GPT-4-Turbo shows minimal improvement (0.7%). This suggests that CoT effectiveness depends heavily on model architecture and training approaches.

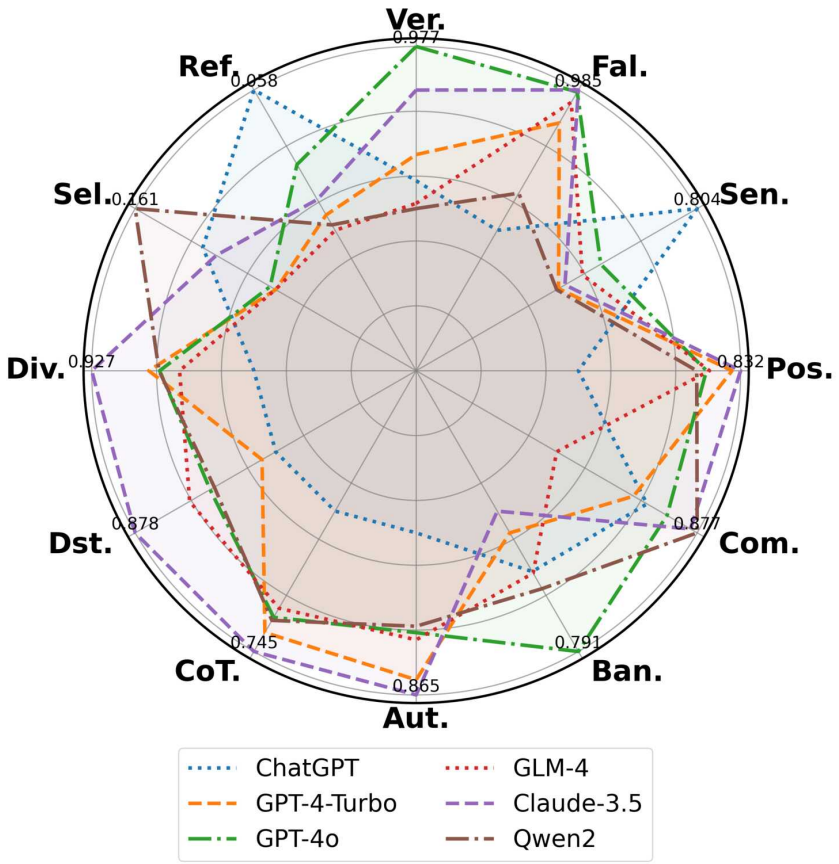

Figure 3: Radar chart comparing robustness rates across all 12 bias types for six LLM judges. Higher values indicate greater resistance to bias. No single model dominates across all bias dimensions.

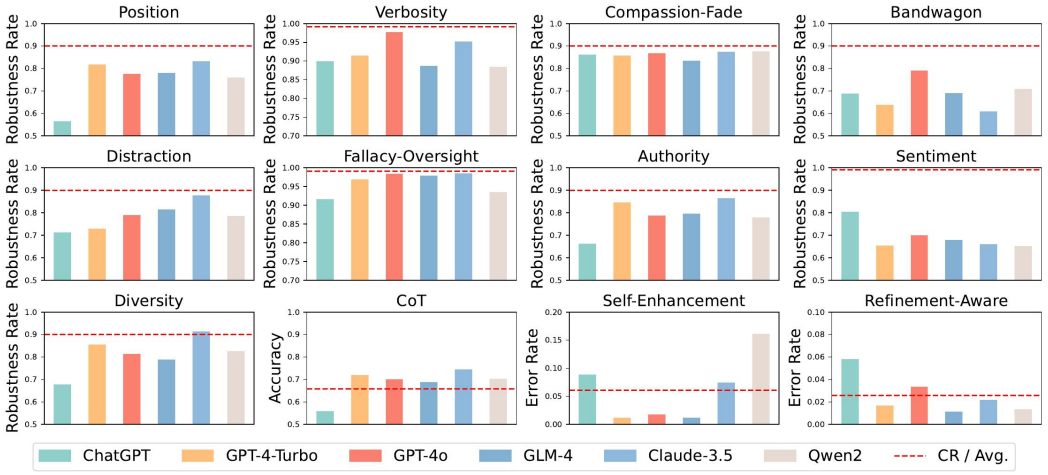

Figure 4: Overall robustness rates for each bias type across all models, with dashed lines indicating consistency rates. Significant variation exists across bias types and model capabilities.

4.3 Position Bias Deep Dive

Position bias proves particularly severe when evaluating multiple answer candidates. Figure 5 shows how robustness rates degrade as the number of candidates increases.

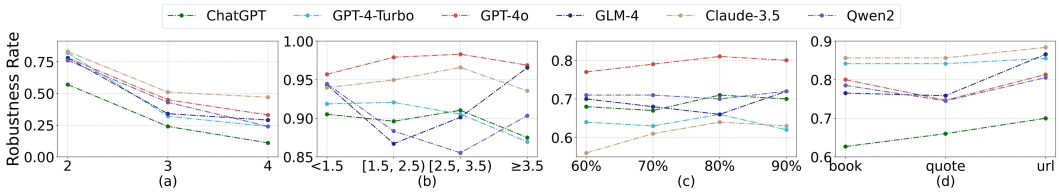

Figure 5: (a) Impact of candidate number on position bias robustness. All models show severe degradation with 3–4 candidates. (b–d) Additional analysis of bias patterns across different contexts and model capabilities.

Most models maintain moderate robustness (0.5–0.8) for 2-candidate comparisons, but performance degrades dramatically to below 0.5 with 3–4 candidates, with critical implications for multi-candidate applications like model selection.

4.4 Self-Enhancement and Refinement-Aware Biases

Table 4 quantifies self-enhancement and refinement-aware biases through error rate analysis.

Self-Enhancement Bias: Error rates vary dramatically across models. Qwen2 shows the highest self-enhancement bias with 16.1% error rate, indicating substantial score inflation when evaluating its own outputs. GPT-4-Turbo demonstrates the lowest bias (1.16%), followed by GLM-4 (1.18%). These findings suggest that model architecture and training procedures significantly influence self-assessment capabilities.

Table 4: Average score and error rate of self-enhancement bias and refinement-aware bias.

Model	Sel. Score↓			Ref. Score↓		
	Self	Other	Error	Ref	+History	Error
ChatGPT	5.21	5.72	8.91	5.23	4.94	5.80
GPT-4-Turbo	6.98	6.90	**1.16**	8.31	8.45	1.66
GPT-4o	7.01	6.89	1.74	7.44	7.20	3.33
GLM-4	7.73	7.64	1.18	7.64	7.73	**1.15**
Claude-3.5	7.04	6.55	7.48	7.51	7.68	2.17
Qwen2	7.64	6.58	16.1	7.29	7.39	1.33

Refinement-Aware Bias: Most models show moderate refinement awareness, with error rates between 1–3%. GLM-4 achieves the lowest refinement bias (1.15%), while GPT-4o shows higher susceptibility (3.33%). Interestingly, some models show slight score decreases when aware of refinement history, suggesting potential over-penalization of revised content.

The heat map in Figure 6 visualizes self-enhancement bias patterns across different model families and evaluation contexts.

4.5 Model-Specific Findings

GPT-4o shows strong content-based bias resistance but social bias vulnerabilities. **Claude-3.5** excels at position/diversity/distraction robustness but struggles with bandwagon effects. **ChatGPT** uniquely resists sentiment bias but shows broad vulnerabilities. **GLM-4 and Qwen2** are competitive overall but exhibit concerning self-enhancement tendencies.

No single model achieves universal robustness, emphasizing the need for ensemble approaches or bias-specific selection.

5 Discussion

5.1 Implications for LLM Evaluation Systems

The persistent bias vulnerabilities across state-of-the-art models require careful mitigation strategies.

Results suggest three mitigation strategies: **multi-model ensembles** leveraging complementary strengths (e.g., Claude-3.5's social robustness with GPT-4o's content resistance), **task-specific selection** (Claude-3.5 for preferences, GPT-4o for facts), and **bias-aware protocols** (randomization for position bias, verification for authority bias).

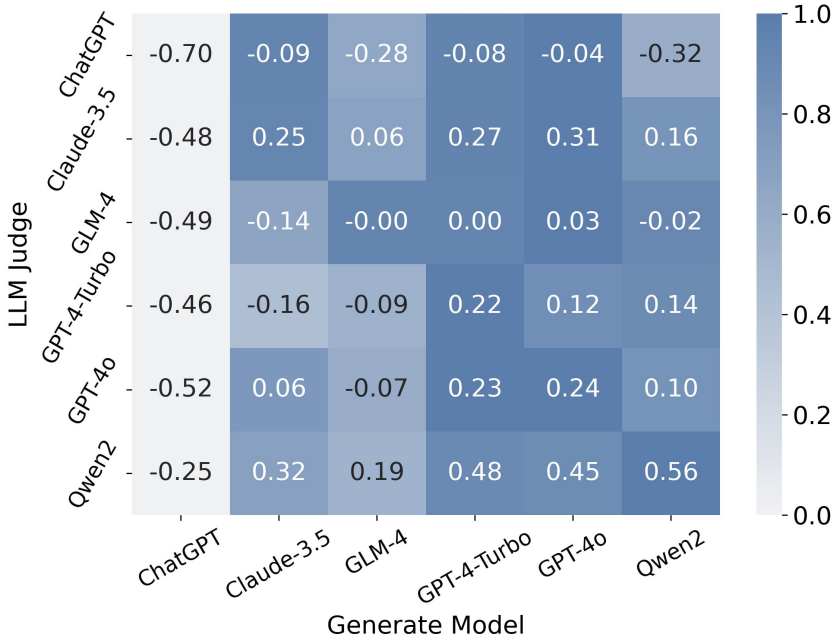

Figure 6: Z-score normalized self-enhancement bias patterns across models. Darker colors indicate stronger self-enhancement tendencies.

5.2 Dataset and Domain Dependencies

Performance varies across datasets: alignment scenarios show heightened social/positional bias vulnerability, fact-related tasks demonstrate better robustness except for sentiment bias, and refinement-aware scenarios reveal inconsistent reactions to revision history.

5.3 Chain-of-Thought Reasoning Effects

CoT prompting shows mixed, model-dependent results: some models benefit from explicit reasoning while others show minimal improvement, with effectiveness varying across bias types.

5.4 Self-Enhancement and Meta-Evaluation Challenges

Substantial self-enhancement biases (error rates exceeding 15%) highlight fundamental self-evaluation challenges, impacting self-improving AI systems and bootstrap evaluation. Refinement-aware bias creates additional challenges in human-AI collaboration workflows.

5.5 Cultural and Architectural Influences

Distinct patterns in GLM-4 and Qwen2 versus Western models suggest cultural/training influences on bias susceptibility, with implications for global deployment and bias-resistant design.

5.6 Limitations and Future Work

Our study has limitations: English-only focus limits multilingual generalizability, principle-guided perturbations may miss real-world bias triggers, and we examine immediate rather than long-term bias accumulation effects. Future work should address multilingual contexts, adversarial approaches, and longitudinal bias studies.

5.7 Broader Impacts and Ethical Considerations

The documented biases have far-reaching implications for AI fairness and safety, potentially perpetuating discrimination and undermining trust in automated assessment. Position bias could systematically disadvantage certain responses, social biases could amplify discrimination against marginalized groups, and self-enhancement bias could create echo chamber effects in self-improving systems. These findings underscore the need for bias-aware design and ongoing monitoring in deployed systems.

6 Conclusion

We have presented CALM, a comprehensive framework for systematically quantifying biases in LLM-as-a-Judge systems. Through our attack-and-detect methodology, we identified and evaluated 12 distinct bias types across three carefully curated datasets, revealing significant vulnerabilities that persist even in state-of-the-art language models.

Key findings show dramatic variation in bias susceptibility across dimensions and models. While models show strong robustness against content-based biases (verbosity, fallacy-oversight), they remain vulnerable to positional and social biases, with position bias particularly concerning (robustness <0.5 for multiple candidates) and self-enhancement bias reaching 16.1% error rates.

CALM establishes a reproducible protocol enabling systematic bias assessment through unified metrics and automated perturbations, providing actionable tools for identifying and mitigating vulnerabilities.

These findings have immediate implications for AI evaluation infrastructure, requiring bias-aware design including ensemble approaches, task-specific selection, and systematic monitoring. Future work should extend CALM to multilingual contexts, investigate mitigation strategies, and examine long-term bias accumulation effects.

As LLM-as-a-Judge systems become central to AI development, understanding and addressing bias vulnerabilities is critical for fair, reliable evaluation. CALM provides a foundation enabling researchers and practitioners to build more robust and equitable AI evaluation systems.

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